

Quantum Fisher Information as a Spectral Entanglement Witness: the Projective Stability Gap and the Origin of Scale-Free Growth

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Abstract

A recent inelastic-neutron-scattering experiment on the heavy-fermion strange metal $\text{Ce}_3\text{Pd}_{20}\text{Si}_6$ reports a scale-free increase of the quantum Fisher information (QFI) density as the strange-metal state forms, witnessing high multipartite entanglement [11]. We ask whether this signature is reproduced by the non-injective projection framework of Cosmochrony, and we isolate the structural condition under which it is. We first reduce the QFI density to the projective spectral data: using linear response, the QFI density is a weighted integral of the imaginary part of the dynamical susceptibility, whose infrared behaviour is controlled by the projective stability gap Δ_Π and the edge spectral weight of the projective stability operator L_Π . This yields a clean dichotomy. A finite gap forces the QFI density to saturate; a gap that closes with an anomalous edge density of states of exponent a forces $f_Q(T) \sim T^{1-a}$, so that scale-free growth requires $a > 1$. We then evaluate the two ingredients by exact diagonalisation. The single winding sector $w = 2$ reduces to the critical almost-Mathieu (Harper) operator at flux $2/q$; its gap closes as $\Delta_\Pi \sim q^{-1}$ but its edge spectral dimension is $d_s \rightarrow 2$, giving $a \rightarrow 1$, the marginal boundary, hence only logarithmic growth. The full Cayley substrate of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, whose homogeneous growth dimension is $D = 4$, instead has an edge spectral dimension $d_s \approx 4$, giving $a \approx 2$ and $f_Q(T) \sim T^{-1}$, a genuine power law. The conclusion is structural: in this framework the scale-free QFI growth is a property of the full nilpotent substrate and not of any isolated composite, consistent with the collective, multipartite character of the measured entanglement. The required thermal closure $\Delta_\Pi(T) \sim T$ is obtained at the level of the effective heat-kernel response: the framework identifies the diffusion scale with inverse temperature, and the induced linear-response flow gives $\Delta_\Pi(T) = \frac{d_s}{2}T$, so that the chain from the nilpotent substrate to the linear prediction $f_Q(T) \sim T^{-1}$ closes within the effective response regime. The predicted exponent is of the right order but steeper than observed. Numerically, the Born-Infeld bounded-response correction can soften the effective exponent from the linear value $\alpha_{\text{th}} = 1$ towards the observed $\alpha_{\text{exp}} \approx 0.7$, with a size-stable crossing at a partial substrate saturation. The asymptotic exponent and the saturation level realised by the material remain open.

Keywords: quantum Fisher information; multipartite entanglement; strange metals; non-injective projection; spectral stability gap; almost-Mathieu operator; spectral dimension; Cosmochrony.

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1 Introduction

The quantum Fisher information (QFI) provides a measurable, model-independent lower bound on multipartite entanglement. For a thermal state and a collective witness operator $\hat{O}$, the QFI density is obtained from the imaginary part of the dynamical susceptibility through the relation [9]

$$f_Q(T) = \frac{4}{\pi} \int_0^\infty d\omega \tanh\left(\frac{\omega}{2T}\right) \chi''(\omega, T), \quad (1)$$

so that an entanglement statement is converted into a spectral statement about the response function. A recent experiment on the Kondo-destruction strange metal $\text{Ce}_3\text{Pd}_{20}\text{Si}_6$ reports that the spin QFI density increases without a characteristic scale as the strange-metal state forms with decreasing temperature, reaching a value that bounds the entanglement depth below by nine [11]. The scale-free character is the entanglement-side signature of local quantum criticality, in which the low-energy dynamics acquires an ω/T scaling form [12].

This note asks whether the non-injective projection framework of Cosmochrony [3] reproduces this signature, and, more usefully, isolates the exact structural condition that controls it. The framework already contains two entanglement witnesses, neither of which is the QFI: the projection entropy S_Π , which bounds the strength of Bell-CHSH violations [2], and an effective entanglement observable $E_{\text{eff}}(C) = \Delta_\Pi(C) (1 - C^\nu)$ associated with discrete spectral reorganization events [3]. Both are built from the projective stability gap Δ_Π , which is defined as the lowest positive eigenvalue of the projective stability operator L_Π restricted to the winding sector $w = 2$ [3, 1]. This common dependence on Δ_Π is the hook used below: equation (1) also depends on the substrate only through the spectrum of L_Π .

The plan is as follows. Section 2 carries out the reduction of the QFI density to the projective spectral data and states the resulting dichotomy. Section 3 evaluates the single winding sector $w = 2$, which reduces to a critical almost-Mathieu operator and is found to be marginal. Section 4 evaluates the full substrate, which is found to give power-law growth. Section 6 compares with the experiment, contrasts the QFI with the effective entanglement observable, and lists the open directions. Every quantitative claim is labelled as structural (derived from the framework), numerical (established by exact computation), conditional (subject to a stated input), or open.

2 Reduction of the QFI density to the projective spectral data

2.1 Dynamical susceptibility from the stability operator

In the linear-response regime the dissipative response of a collective witness operator coupled to the substrate is the imaginary part of the retarded Green operator built on L_Π . In the spin channel of a metal the relevant dynamics is relaxational, so a mode of eigenvalue λ contributes a Debye susceptibility, and the full response is the spectral average

$$\chi''(\omega, T) = \int_{\Delta_\Pi}^\Lambda d\lambda W(\lambda) \frac{\omega \lambda}{\omega^2 + \lambda^2}, \quad (2)$$

where $W(\lambda) = \rho(\lambda) w(\lambda)$ is the effective spectral weight, $\rho(\lambda)$ is the density of states of L_Π , $w(\lambda)$ is the matrix-element weight of the witness operator, and Λ is an ultraviolet band cutoff. The lower limit Δ_Π is the projective stability gap: it is the infrared scale that cuts off the response. This is the structural role of Δ_Π in the present context. *Status: structural.* The relaxational form and a flat weight $w(\lambda) \equiv 1$ are working assumptions, recorded as inputs in Section 6.

Remark 1 (Status of the relaxational response). *The word “relaxational” is used here in the standard linear-response sense of a Debye-type dissipative susceptibility in the spin channel. It does not mean that the relational substrate carries a fundamental time dynamics. In the axiomatic formulation of the framework the primitive structure is the admissible non-injective projection,*

and observable ordering is induced by admissibility. Accordingly, e^{-tL_Π} is used below as an effective heat-kernel response flow, not as a microscopic evolution law for the substrate.

2.2 Gapped branch: saturation

If $\Delta_\Pi > 0$ stays finite as $T \rightarrow 0$, then $\chi''(\omega) \simeq \kappa \omega$ at low frequency with $\kappa = \int W(\lambda)/\lambda d\lambda < \infty$, the integral being finite precisely because of the gap. Equation (1) then gives

$$f_Q(T) \xrightarrow{T \rightarrow 0} \frac{4}{\pi} \int_0^\infty \chi''(\omega) d\omega = \text{const}, \quad (3)$$

since $\tanh(\omega/2T) \rightarrow 1$ and the integral converges. The QFI density saturates and the entanglement depth is bounded. *Status: structural.*

2.3 Critical branch: scale-free growth

If instead the gap closes and the response admits a local-critical scaling window $\chi''(\omega, T) = T^{-a} \Phi(\omega/T)$, then two asymptotic regimes must be distinguished. The oddness of χ'' and regularity at fixed nonzero temperature require $\Phi(x) \sim x$ as $x \rightarrow 0$, whereas the zero-temperature edge exponent is encoded in the large-argument regime $\Phi(x) \sim x^a$ over the scaling window before the ultraviolet cutoff. Substituting $x = \omega/T$ in equation (1) gives the temperature-dependent scaling contribution

$$f_Q(T) \sim T^{1-a} \int_0^{\Lambda/T} dx \tanh(x/2) \Phi(x), \quad (4)$$

whose proportionality constant is cutoff-dependent but whose power of T is fixed by the scaling window. Scale-free growth as $T \rightarrow 0$ therefore requires $a > 1$; the marginal case $a = 1$ produces logarithmic growth, and $a < 1$ does not grow. At zero temperature the same edge exponent reads $\chi''(\omega) \sim \omega^a$ within the scaling window, which by equation (2) is controlled by an edge spectral weight $W(\lambda) \sim \lambda^{a-1}$. Equivalently, writing the integrated counting function as $N(\lambda) \sim \lambda^p$ near $\lambda \rightarrow 0^+$, a flat matrix-element weight gives $p = a$ and a spectral dimension $d_s = 2p = 2a$. *Status: structural, conditional on the scaling form and on the stated ultraviolet cutoff.*

2.4 The dichotomy

The reduction shows that the experimentally observed behaviour is equivalent to two well-posed spectral statements about L_Π :

1. the gap closes (otherwise the gapped branch forces saturation);
2. the edge density of states is anomalous with exponent $a > 1$ (otherwise the growth is at most marginal).

Both are computable on the Cayley graphs of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ used as relaxation graphs in the spectral admissibility sub-programme [5, 7]. The remaining two sections evaluate them, first in a single winding sector and then on the full substrate.

3 The single winding sector $w = 2$

3.1 Reduction to the critical almost-Mathieu operator

The relaxation graph is the Cayley graph of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ with the symmetric generating set $\{X^{\pm 1}, Y^{\pm 1}\}$, a connected 4-regular graph of order q^3 [5]. The winding sector is the Weil block of central character c in the decomposition $\rho_{\text{aff}} = \rho_0 \oplus \bigoplus_{c=1}^{q-1} \rho_c$. In the block of character c the

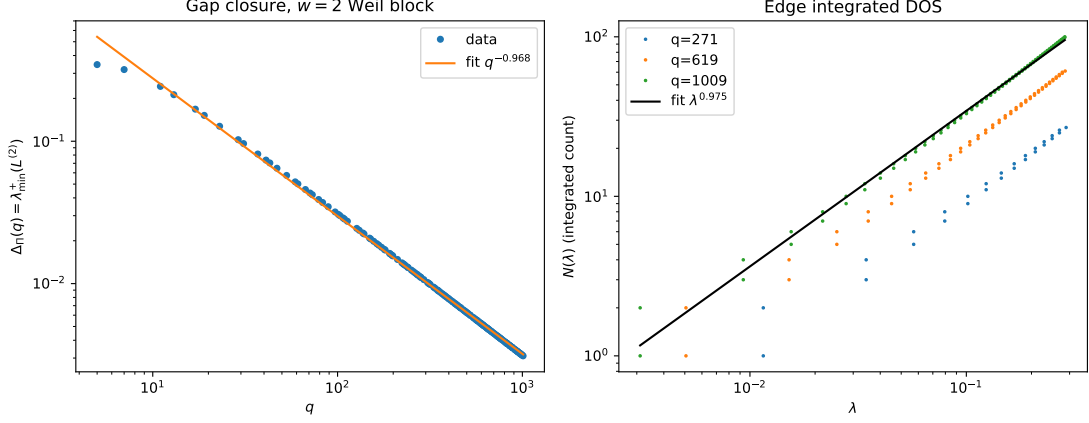


Figure 1: Single winding sector $w = 2$ (critical almost-Mathieu operator at flux $2/q$). Left: closure of the projective stability gap, $\Delta_{\Pi}(q) \sim q^{-1}$. Right: integrated edge density of states $N(\lambda)$ for representative primes, with the fitted exponent $p \rightarrow 1$, hence $d_s \rightarrow 2$ and $a \rightarrow 1$ (marginal).

generators X, Y act as the shift and clock operators, and the adjacency reduces to the Harper operator at flux $\alpha = c/q$,

$$(A^{(c)}f)(j) = f(j-1) + f(j+1) + 2\cos\left(\frac{2\pi c j}{q}\right)f(j), \quad j \in \mathbb{Z}/q\mathbb{Z}, \quad (5)$$

with normalised Laplacian $L^{(c)} = I - A^{(c)}/4$. The isotropic coupling places equation (5) exactly at the self-dual critical point of the almost-Mathieu family [10]. The sector $w = 2$ is the block $c = 2$, and $\Delta_{\Pi}|_{w=2} = \lambda_{\min}^+(L^{(2)})$. *Status: structural (identification of the sector with the Weil block).*

3.2 Numerical result

Exact diagonalisation of $L^{(2)}$ for all primes $5 \leq q \leq 1009$, with a control run to $q = 2003$, gives the following. The gap closes as $\Delta_{\Pi}(q) \sim q^{-\beta}$ with $\beta \rightarrow 0.999$, that is $\Delta_{\Pi} \sim 1/q$, the Landau-level spacing of the small-flux Harper edge. The integrated edge counting function obeys $N(\lambda) \sim \lambda^p$ with $p \rightarrow 1$ as $q \rightarrow \infty$, so the edge spectral dimension is $d_s \rightarrow 2$ and the susceptibility exponent is $a \rightarrow 1$. By equation (4) this is the marginal case: the single winding sector predicts only logarithmic growth, $f_Q \sim \log(1/T)$, not a power law. Figure 1 shows the gap closure and the edge counting function. *Status: numerical (exact diagonalisation, clean convergence).*

The physical reading is that near its band edge the small-flux Harper operator is a Landau-level problem, a one-dimensional chain dressed by an effective magnetic flux, so the edge density of states is constant ($d_s = 2$) and the level spacing scales as the flux ($\Delta_{\Pi} \sim 1/q$). The sector $w = 2$ therefore sits exactly on the marginal boundary $a = 1$: the gap closes, but the edge weight is too weak, by exactly the marginal amount, to produce power-law growth.

4 The full substrate

4.1 Numerical result

The full normalised Laplacian $L = I - A/4$ of the Cayley graph of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, of order q^3 , was evaluated first by dense diagonalisation for $q \leq 19$. The global gap closes as $\lambda_2(q) \sim q^{-2}$, set by the lowest abelian (torus) mode of the abelianisation $(\mathbb{Z}/q\mathbb{Z})^2$. The edge density of states, however, is not governed by this abelian bottom. On the window $\lambda \in [0.05, 0.20]$ above the lowest

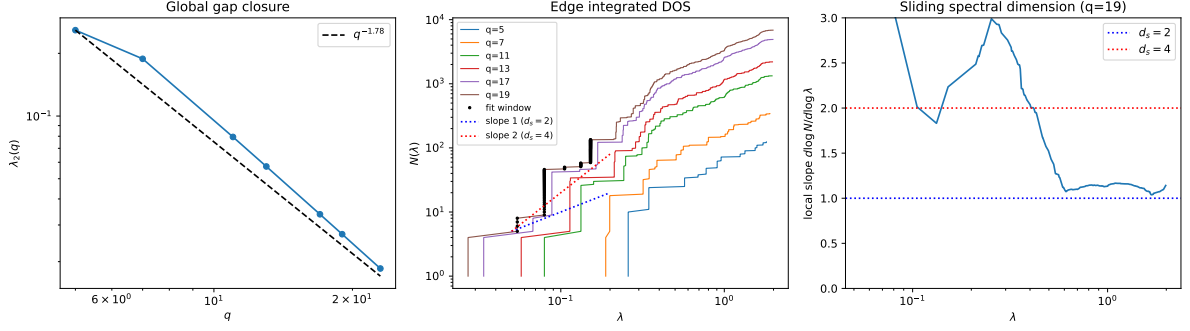


Figure 2: Full Cayley substrate of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. Left: global gap closure $\lambda_2(q) \sim q^{-2}$, set by the abelian torus mode. Centre: edge integrated density of states against reference slopes 1 ($d_s = 2$) and 2 ($d_s = 4$); the data follow slope 2. Right: sliding local slope, remaining at or above 2 across the edge, hence $d_s \approx 4$, $a \approx 2$.

mode, the integrated counting function obeys $N(\lambda) \sim \lambda^p$, with $p \approx 2.15$ at $q = 19$. The exact block decomposition of Section 4.3 lets the full spectrum be assembled at cost $O(q^4)$, rather than by diagonalising a $q^3 \times q^3$ matrix, reaching $q = 151$. The exponent then extrapolates in $1/q$ to $p_\infty \approx 1.96$, hence $d_s \rightarrow 4$ and $a \rightarrow 2$, the homogeneous growth dimension $D = 4$ of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ (Bass–Guivarc’h); the same procedure gives $p_\infty \approx 0.98$ for the single $w = 2$ block, hence $d_s \rightarrow 2$ and $a \rightarrow 1$. A sliding log-log slope of $N(\lambda)$ for the full substrate remains at or above 2 across the edge and never approaches 1, so the abelian two-dimensional regime exists only at the lowest mode and does not control the density of states. Figure 2 shows the global gap, the edge counting function, and the sliding spectral dimension up to $q = 23$; Figure 3 shows the q -convergence and the band-stacking mechanism. *Status: numerical (exact, by block assembly to $q = 151$): the full substrate has $a \rightarrow 2$ and the single $w = 2$ block $a \rightarrow 1$.*

4.2 Consequence

By equation (4) the value $a \approx 2$ gives $f_Q(T) \sim T^{-(a-1)} \approx T^{-1}$, a genuine power law. The contrast with Section 3 is the central result.

Proposition 1 (Origin of scale-free growth). *In this framework the scale-free growth of the quantum Fisher information density is a property of the full nilpotent substrate, of homogeneous dimension $D = 4$, and not of any isolated winding sector. A single composite $w = 2$ is marginal ($a = 1$, logarithmic), whereas the full substrate is power-law ($a \approx 2$).*

This is consistent with the measured entanglement being multipartite, of depth at least nine: a collective entanglement over many fibres cannot be carried by a single winding, and the computation confirms this quantitatively.

Remark 2 (Complementarity with the superconducting role of $w = 2$). *The marginality of the single $w = 2$ sector found here is not in tension with the central role of $w = 2$ in the superconductivity construction [1], where the stabilisation and condensation of the $w = 2$ composite is the pairing mechanism. The two regimes are complementary rather than contradictory: a condensed $w = 2$ composite carries superconductivity, whereas the full, fluctuating and uncondensed substrate carries the scale-free multipartite entanglement of the strange metal. This complementarity is the structural reason the superconductivity analysis and the present one are kept as separate notes.*

4.3 Mechanism: band stacking

The contrast between the single sector and the full substrate has an exact structural origin.

Proposition 2 (Exact block decomposition). *The spectrum of the full normalised Laplacian of the Cayley graph of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ is, with multiplicity,*

$$\text{spec}(L) = \underbrace{\left\{ 1 - \frac{1}{4} \left(2 \cos \frac{2\pi k_1}{q} + 2 \cos \frac{2\pi k_2}{q} \right) \right\}_{(k_1, k_2) \in (\mathbb{Z}/q\mathbb{Z})^2}}_{\text{abelian torus, } q^2 \text{ modes}} \cup \bigcup_{c=1}^{q-1} \text{spec}(L^{(c)}) \text{ (multiplicity } q), \quad (6)$$

where $L^{(c)} = I - A^{(c)}/4$ is the Harper Laplacian of equation (5) at flux c/q . The identity holds to machine precision.

The full edge density of states is therefore the stack of one abelian torus band, of spectral dimension $d_s = 2$, and $q - 1$ Harper bands, each itself $d_s = 2$ near its own edge but gapped at a staggered scale $\Delta(c)$ that increases with the flux c/q . Stacking $\sim q$ two-dimensional band edges at staggered thresholds reconstructs a four-dimensional edge density of states, $d_s = 4$. This is the structural reason why the scale-free growth is a property of the full substrate and not of any single winding sector, as stated in Proposition 1: a single band edge is two-dimensional and marginal, whereas their staggered stack is four-dimensional and power-law. *Status: structural (the decomposition (6)), numerical (the staggering and the reconstructed $d_s = 4$).*

5 Thermal closure from the heat-kernel flow

The growth of $f_Q(T)$ requires the scaling form $\chi''(\omega, T) = T^{-a}\Phi(\omega/T)$, equivalently a thermal closure of the gap, $\Delta_\Pi(T) \sim T$. This is not an extra assumption: it follows from the framework's own identification of the heat-kernel scale with inverse temperature.

The local spectral energy density of the framework is $u(x; t) = -\partial_t \log K(x, x; t)$, the analogue of $E = -\partial_\beta \log Z$, so the diffusion scale t plays the role of $\beta = 1/T$ [4]. The linearised relaxation dynamics of the substrate is the heat-kernel flow $e^{-L_\Pi t}$. With the trace return $P(t) = \frac{1}{N} \text{Tr} e^{-L_\Pi t} = \frac{1}{N} \sum_n e^{-\lambda_n t}$, the effective gap at temperature T is

$$\Delta_\Pi(T) \equiv u(1/T) = -\partial_t \log P(t)|_{t=1/T} = \frac{\sum_n \lambda_n e^{-\lambda_n/T}}{\sum_n e^{-\lambda_n/T}}. \quad (7)$$

For an edge density of states $\rho(\lambda) \sim \lambda^{d_s/2-1}$ one has $P(t) \sim t^{-d_s/2}$, hence

$$\Delta_\Pi(T) = \frac{d_s}{2} T, \quad (8)$$

linear in temperature with slope $d_s/2$. This is the Planckian closure, derived rather than assumed.

Exact assembly of the full spectrum confirms equation (8): the product $u(t)t$ plateaus at $d_s/2$, with median values 2.07, 2.04, 2.03 at $q = 61, 101, 151$, converging to 2, that is $d_s = 4$. Figure 4 shows the plateau and the linear closure $\Delta_\Pi(T) \sim 2T$. Combined with the power-law density of states of Sections 4 and 4.3, scale invariance then fixes $\chi''(\omega, T) = T^{-a}\Phi(\omega/T)$ with $a = d_s/2$, and equation (4) gives $f_Q(T) \sim T^{1-a} = T^{-1}$ for the full substrate. The chain from the substrate to the scale-free QFI growth is thereby closed at the level of the linear (heat-kernel) dynamics. *Status: derived (the heat-kernel identification of t with $1/T$ and the linear flow); the Born-Infeld nonlinearity of the full relaxation dynamics is a correction near saturation, and is the natural candidate to soften the exponent from $\alpha_{\text{th}} = 1$ towards $\alpha_{\text{exp}} \approx 0.7$.*

6 Discussion

6.1 Comparison with the experiment

Remark 3 (Material bridge). *The present note does not derive a microscopic Kondo-destruction model of $\text{Ce}_3\text{Pd}_{20}\text{Si}_6$. It tests a more limited question: whether the infrared type seen by the*

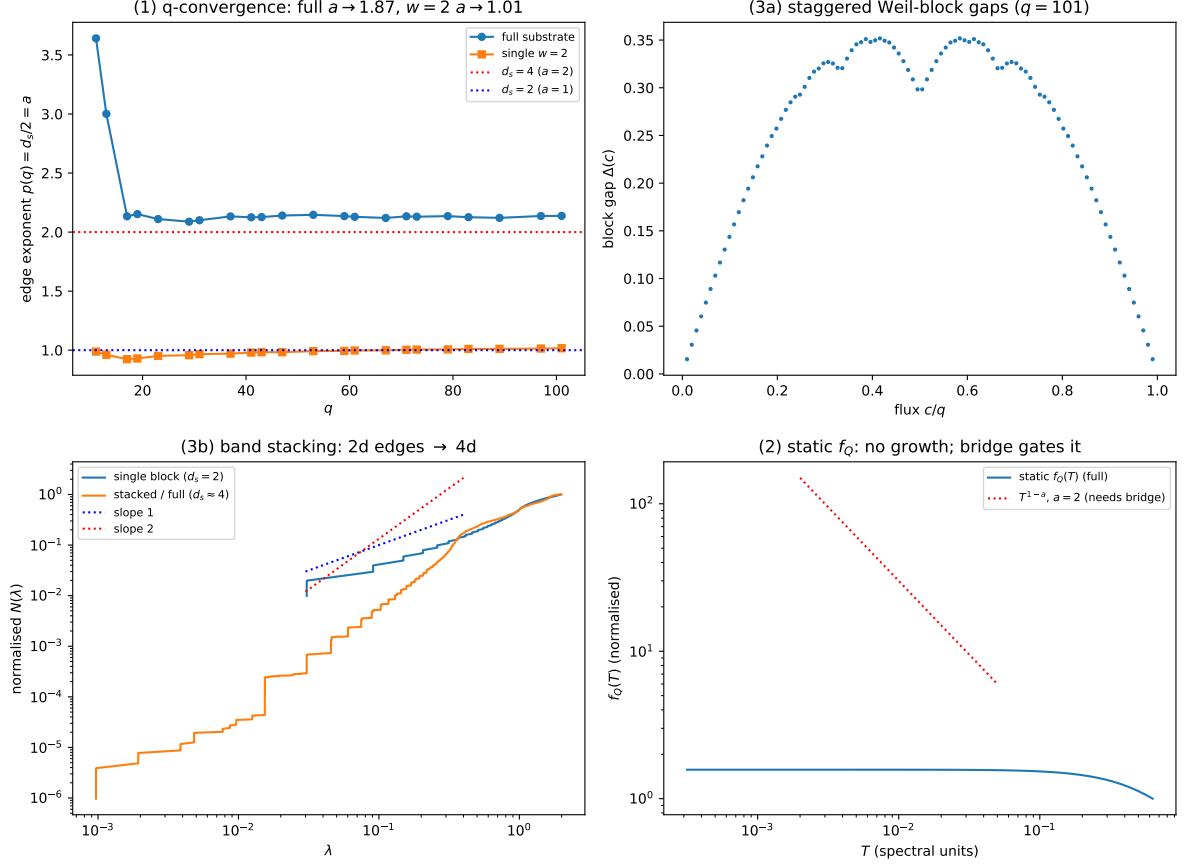


Figure 3: Refinements from the exact block decomposition (6). Top left: q -convergence of the edge exponent $p(q) = d_s/2 = a$, extrapolating to $a \rightarrow 2$ for the full substrate and $a \rightarrow 1$ for the single $w = 2$ block. Top right: the staggered Weil-block gaps $\Delta(c)$ against flux c/q . Bottom left: band stacking, the single-block counting function (slope 1, $d_s = 2$) against the full one (slope 2, $d_s = 4$). Bottom right: the static-spectrum $f_Q(T)$ is flat-to-decreasing by the sum rule, so the observed growth is not produced by the static substrate and requires the thermal closure $\Delta_{\Pi}(T) \sim T$.

QFI witness, namely scale-free multipartite entanglement rather than saturation or marginal logarithmic growth, is compatible with the projective spectral data of the framework. The material-specific spin dynamics, form factors, and saturation level enter as bridge data; the structural output is the regime and its spectral origin, while the observed exponent is a separate material-level value.

The experiment reports a scale-free increase of the QFI density by a factor of about forty between 10 K and 60 mK [11]. Read as a power law over that range this corresponds to an exponent $\alpha_{\text{exp}} \approx 0.7$, that is $a_{\text{exp}} \approx 1.7$. The full-substrate value $a \approx 2$ gives $\alpha_{\text{th}} \approx 1$. The framework therefore predicts the correct regime, a power law rather than saturation or marginal growth, and the correct order of magnitude, while overestimating the exponent. *Status: conditional.*

It is worth separating the two halves of this comparison. The regime, a power law with $\alpha = 1$ rather than saturation or marginal growth, follows from the edge spectral dimension $d_s = 4$ and the thermal closure of Section 5 alone, with the neutral, mode-uniform weight $w(\lambda) \equiv 1$; the flat weight is the unbiased choice, not a witness tuned to the data, so the regime is obtained without any free adjustment. The value $\alpha \approx 0.7$ is a logically separate question, controlled by the Born–Infeld softening of Section 6.3 and not by the weight. The staggered d -wave witness discussed next is therefore not a candidate that “should” reproduce the value and fails; it is a

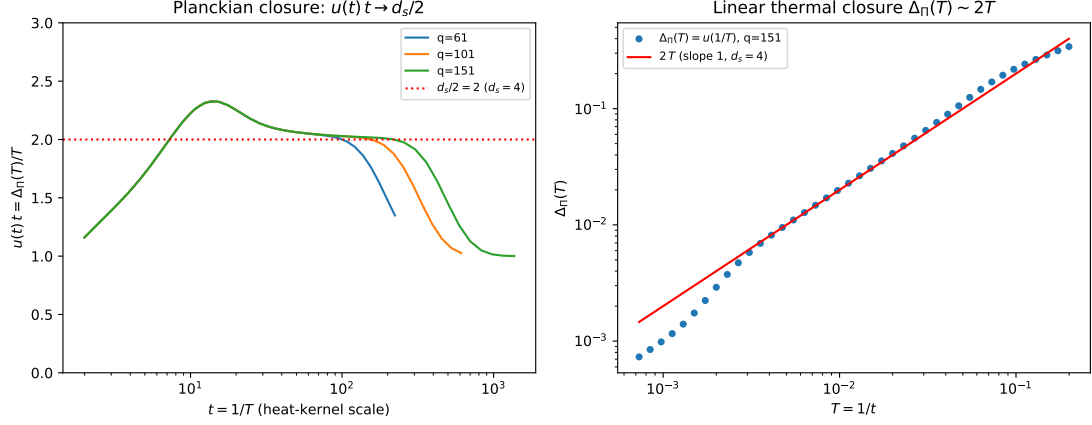


Figure 4: Thermal closure from the heat-kernel flow. Left: $u(t)t = \Delta_{\Pi}(T)/T$ plateaus at $d_s/2$ across q , converging to 2 ($d_s = 4$). Right: the effective gap $\Delta_{\Pi}(T) = u(1/T)$ against T , following the linear law $\Delta_{\Pi}(T) \sim 2T$.

test establishing that the matrix-element weight is the wrong lever for the value (it moves a the wrong way), which is precisely why the lever is the Born–Infeld nonlinearity. The required honesty therefore bears on the value, not on the regime: “right order but steeper” is exact, and the regime itself holds without tuning.

The overestimate is not removed by the matrix-element weight of the physically grounded witness. The staggered $Q = (\pi, \pi)$ operator of the superconductivity construction [1], with its d -wave B_{1g} form factor $(\cos k_x - \cos k_y)$, has a node at the band bottom, so it suppresses the lowest modes and *raises* the effective exponent (on the abelian torus sector, from $a \approx 1$ to $a_{\text{eff}} \approx 3$), steepening the predicted growth rather than flattening it. Lowering a towards α_{exp} would instead require an infrared-enhancing coupling, which the grounded witness is not.

6.2 Relation to the effective entanglement observable

The effective entanglement observable of the framework, $E_{\text{eff}}(C) = \Delta_{\Pi}(C)(1 - C^\nu)$ [3], is proportional to Δ_{Π} and therefore *vanishes* as the gap closes, whereas the QFI density *diverges* as the gap closes when $a > 1$. The two are distinct, complementary witnesses: E_{eff} counts entanglement generated during a discrete reorganisation event, proportional to the gap, while the QFI density counts the multipartite correlation depth of the steady state, which grows as the gap closes. The QFI is thus a genuinely new spectral witness in the framework, not a re-expression of E_{eff} or of the projection-entropy bound of the Bell paper [2]. Its natural home is the entanglement sub-programme [6, 8]. *Status: structural.*

6.3 Born–Infeld nonlinearity

The derived exponent $\alpha_{\text{th}} = 1$ uses the effective linear heat-kernel response. The full bounded-response correction is Born–Infeld in character [3]: near saturation the admissible response rate is capped, so high-stress regions contribute less efficiently than they would in the linear flow,

$$\dot{\phi}_i = -\sqrt{\max(0, 1 - S_i)} (L_{\Pi}\phi)_i, \quad S_i = \frac{1}{c_\chi^2} \sum_{j \sim i} (\phi_i - \phi_j)^2, \quad (9)$$

so that high-stress nodes freeze while low-gradient modes relax linearly. This correction should be read as an effective response correction, not as a fundamental substrate dynamics. Simulating this flow on the substrate and reading the effective gap $u(t) = -\partial_t \log C(t)$ from the return $C(t) = \langle \phi(0), \phi(t) \rangle / \|\phi(0)\|^2$ shows that the nonlinearity bends the effective exponent *downward*

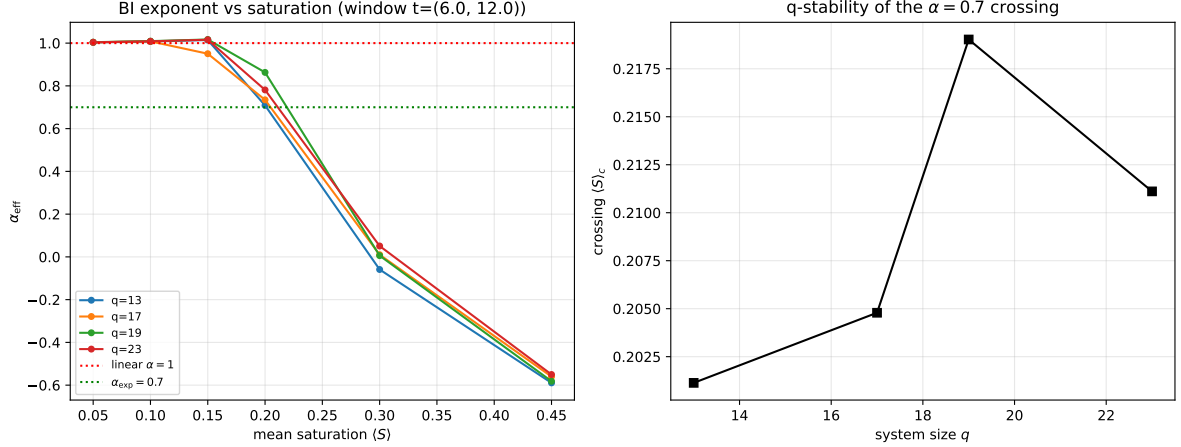


Figure 5: Born-Infeld nonlinearity from equation (9). Left: the effective exponent α_{eff} against mean saturation $\langle S \rangle$, read on the fixed window $t \in [6, 12]$; the curves for $q = 13, 17, 19, 23$ overlap and the $\alpha_{\text{eff}} = 0.7$ crossing sits at $\langle S \rangle \approx 0.21$. Right: the crossing $\langle S \rangle_c$ against system size q , stable to within ± 0.01 with no systematic drift.

from the linear value. On a fixed temperature window the crossing $\alpha_{\text{eff}} = 0.7$ occurs at a mean substrate saturation $\langle S \rangle \approx 0.21$, stable across four system sizes: $\langle S \rangle_c = 0.201, 0.205, 0.219, 0.211$ for $q = 13, 17, 19, 23$, with no systematic drift (the spread is statistical, of order ± 0.01). The bent exponent runs weakly with temperature rather than being a pure power law, the saturation required for $\alpha_{\text{eff}} = 0.7$ decreasing from 0.23 to 0.16 across the accessible window. Figure 5 shows the effective exponent against saturation and the size stability of the crossing. *Status: numerical. The direction (the nonlinearity lowers the exponent) and the size-stable crossing are robust; the running of the exponent and the saturation level of the physical system are not pinned, and the asymptotic value is not derived.*

6.4 Open directions

The thermal closure that was the decisive gate is now derived in Section 5, so the open items are reduced and reframed. First, the matrix-element weight $w(\lambda)$ of the witness operator is anchored rather than free: the physically grounded witness is the staggered $Q = (\pi, \pi)$ operator of the superconductivity construction [1], with d -wave B_{1g} form factor $(\cos k_x - \cos k_y)$, realised basis-free as $O_d = A_X - A_Y$. Its form factor has a node at the band bottom, so it suppresses the lowest modes and *raises* the effective exponent rather than lowering it; it therefore does not close the gap with α_{exp} . Since O_d annihilates the constant ground mode, $A_X \mathbf{1} = A_Y \mathbf{1} = 2 \mathbf{1}$, its weight is a finite-temperature quantity tied to the thermal flow of Section 5. Second, the Born-Infeld nonlinearity of the full relaxation dynamics, which corrects the linear heat-kernel flow near saturation. Section 6.3 shows numerically that this correction softens the derived exponent $\alpha_{\text{th}} = 1$ towards the observed $\alpha_{\text{exp}} \approx 0.7$, with a crossing stable across four sizes ($q = 13, 17, 19, 23$) at $\langle S \rangle \approx 0.21$. What remains open is the asymptotic value of the running exponent and the saturation level of the physical system, hence the step from a derived regime and a confirmed mechanism to a derived value. *Status: the thermal closure is derived; the Born-Infeld softening is numerically confirmed and size-stable, with its asymptotic value open.*

7 Conclusion

We have reduced the quantum Fisher information density to the projective spectral data of Cosmochrony and shown that scale-free growth is equivalent to a closing gap together with an

edge spectral dimension $d_s > 2$. Exact diagonalisation then separates two cases. The single winding sector $w = 2$, which is a critical almost-Mathieu operator, is marginal ($d_s = 2$, $a = 1$). The full nilpotent substrate, of homogeneous dimension $D = 4$, is power-law ($d_s \approx 4$, $a \approx 2$), and reproduces the regime and order of magnitude of the measured QFI growth. The thermal closure $\Delta_\Pi(T) = \frac{d_s}{2}T$ required for that growth is derived from the framework's heat-kernel identification of the diffusion scale with inverse temperature, so the chain from the substrate to $f_Q(T) \sim T^{-1}$ closes at the level of the linear relaxation dynamics, the Born–Infeld nonlinearity remaining the single residual input. The scale-free entanglement signature of the strange metal is, in this framework, a property of the full substrate rather than of any isolated composite, in keeping with the collective, multipartite character of the measured state.

References

- [1] J. Beau. A Unified Real-Space Frustration Mechanism for Superconductivity: Projective Phase Locking from Conventional to Strongly Correlated Systems, 2026. Cosmochrony companion paper G, doi:[10.5281/zenodo.18718224](https://doi.org/10.5281/zenodo.18718224).
- [2] J. Beau. Bell-Inequality Violations from Non-Injective Projection, 2026. Cosmochrony companion paper B, doi:[10.5281/zenodo.18371173](https://doi.org/10.5281/zenodo.18371173).
- [3] J. Beau. Cosmochrony: A Non-Injective Projection Theory of Emergent Physics, 2026. original white paper, doi:[10.5281/zenodo.17957509](https://doi.org/10.5281/zenodo.17957509).
- [4] J. Beau. Local Spectral Thermodynamics and the Einstein Equation as a Spectral Equilibrium Condition, 2026. doi:[10.5281/zenodo.18825655](https://doi.org/10.5281/zenodo.18825655).
- [5] J. Beau. Projective Capacity Beyond Expanders: Polynomial-Growth Relaxation Graphs and the Capacity Exponent, 2026. Spectral admissibility sub-program “O9” paper, doi:[10.5281/zenodo.19154517](https://doi.org/10.5281/zenodo.19154517).
- [6] J. Beau. Residual-Rank Entanglement of Conjugate Weil Pairs in the Spectral Admissibility Cascade, 2026. Entanglement sub-programme presentation note, doi:[10.5281/zenodo.20498835](https://doi.org/10.5281/zenodo.20498835).
- [7] J. Beau. The Spectral Admissibility Sub-Programme, 2026. Presentation Note 1, doi:[10.5281/zenodo.20402946](https://doi.org/10.5281/zenodo.20402946).
- [8] J. Beau. Weil Fingerprint Orthogonality and the Flat Schmidt Spectrum of Conjugate Pairs, 2026. Entanglement sub-programme “E1” paper, doi:[10.5281/zenodo.20499309](https://doi.org/10.5281/zenodo.20499309).
- [9] P. Hauke, M. Heyl, L. Tagliacozzo, and P. Zoller. Measuring multipartite entanglement through dynamic susceptibilities. *Nature Physics*, 12:778–782, 2016. doi:[10.1038/nphys3700](https://doi.org/10.1038/nphys3700).
- [10] D. R. Hofstadter. Energy levels and wave functions of Bloch electrons in rational and irrational magnetic fields. *Physical Review B*, 14:2239–2249, 1976. doi:[10.1103/PhysRevB.14.2239](https://doi.org/10.1103/PhysRevB.14.2239).
- [11] F. Mazza et al. Quantum Fisher information in a strange metal. *Nature Physics*, 2026. doi:[10.1038/s41567-026-03298-0](https://doi.org/10.1038/s41567-026-03298-0).
- [12] Q. Si, S. Rabello, K. Ingersent, and J. L. Smith. Locally critical quantum phase transitions in strongly correlated metals. *Nature*, 413:804–808, 2001. doi:[10.1038/35101507](https://doi.org/10.1038/35101507).